

STA 35C: Statistical Data Science III

Lecture 20: Basis Functions & Regression Splines

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Announcements

Midterm 2 solutions and scores are posted online

- You can review your graded exam during tomorrow's discussion section

Regrade requests

- If you believe a score should be adjusted, please email the TA **by noon on Wednesday, May 20** with:
 - The specific problem(s) you request regrading
 - A clear explanation of why you believe your answer merits more credit (e.g., by pointing out the key elements in your answer that match the official solution)

Where we are now

So far, we have studied foundational topics in statistical learning:

- **Regression and Classification** basics
- **Resampling methods:** Cross-validation & bootstrap
- **Model selection:** Subset selection and regularization (ridge & lasso)
- **Multiple testing:** FP inflation, FWER and FDR controls

Now we move to two final themes for the rest of the course:

- **Beyond linear models:**
 - Basis functions & regression splines
- **Unsupervised learning:** structure in X without a response Y
 - Principal component analysis (PCA)
 - Clustering

Agenda

Today's topic: We continue the model-flexibility story. Instead of only selecting or shrinking existing predictors, we create richer transformed features using basis functions.

- **Basis functions**
 - Recall: polynomial regression
 - Step functions
 - How basis functions unify these ideas
- **Regression splines**
 - Piecewise polynomials
 - Smoothness constraints at knots
 - Truncated power basis representation
 - Preview: boundary behaviors and "natural" splines

Motivation for basis functions: Beyond linear models

Challenge: Linear models are powerful but can sometimes be too restrictive

- Assumes $Y \approx \beta_0 + \sum_{j=1}^p \beta_j X_j$, a linear function of the raw predictors
- Real data often exhibits more complex, nonlinear relationships

Goal: Capture nonlinear structure while keeping estimation as tractable as least squares

Examples:

- *Polynomial regression:* use $(X, X^2, X^3, \dots)$
- *Step functions:* approximate the regression function by piecewise-constant segments

Today's plan:

- Review polynomial regression & define step functions
- Unify these via *basis functions*
- Introduce *splines* for flexible piecewise-defined models

Example 1: Polynomial regression

Polynomial regression: Replace the standard linear model

$$Y = \beta_0 + \beta_1 X + \epsilon$$

with a polynomial:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_d X^d + \epsilon$$

Remarks:

- Although nonlinear in X , the model is linear in the coefficients β_j
 - Treat $X, X^2, \dots, X^d$ as transformed predictors
 - The coefficients $\beta_1, \dots, \beta_d$ can be estimated by standard least squares
- Typically, a moderate degree d (2–4) is used to avoid overfitting

Example 1: Polynomial regression (illustration)

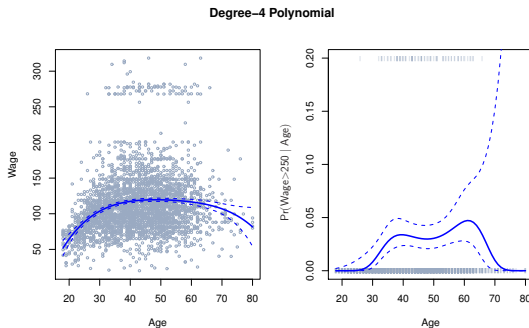


Figure: Polynomial fit on Wage data. (Left) A degree-4 polynomial of wage on age, with 95% confidence bands. (Right) Logistic regression for wage > \$250k using a degree-4 polynomial [JWHT21, Figure 7.1].

- **Advantage:** More flexible than a strictly linear model
- **Limitation:** A polynomial is global; changing the fit in one region can affect the entire curve, especially wildly near boundaries

Example 2: Step functions

Idea: Partition a continuous variable X into intervals, treating each as a separate "level"

- For cutpoints $c_1 < c_2 < c_3 \dots$, define indicator functions:

$$C_0(X) := I_{(-\infty, c_1]}(X), \quad C_1(X) := I_{(c_1, c_2]}(X), \quad C_2(X) := I_{(c_2, c_3]}(X) \dots$$

- The indicator function

$$I_S(x) = \begin{cases} 1 & \text{if } x \in S, \\ 0 & \text{if } x \notin S. \end{cases}$$

- Equivalently, we often write $I(x \in S)$ for $I_S(x)$
- The fitted model is piecewise constant:

$$\mathbb{E}[Y \mid X = x] = \beta_0 + \beta_1 \cdot C_1(x) + \beta_2 \cdot C_2(x) + \dots = \begin{cases} \beta_0 & x \leq c_1, \\ \beta_0 + \beta_1 & c_1 < x \leq c_2, \\ \beta_0 + \beta_2 & c_2 < x \leq c_3, \\ \vdots & \vdots \end{cases}$$

Example 2: Step functions (illustration)

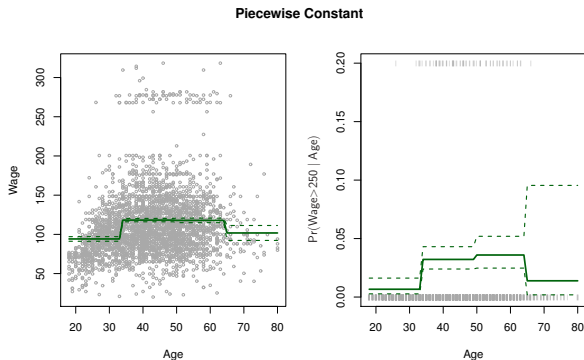


Figure: Step-function fit on Wage data. (Left) Piecewise-constant model for wage on age, with 95% confidence bands. (Right) Logistic regression for wage > \$250k using a step function [JWHT21, Figure 7.2].

- **Advantage:** Easy to capture abrupt changes (“jumps”)
- **Limitation:** Not smooth; can be too coarse with few cutpoints

Basis functions: Bridging polynomial, step, and more

Key idea: Transform the raw predictor X to construct new features $\{b_1(X), \dots, b_K(X)\}$, then fit a linear model in those features:

$$Y \approx \beta_0 + \beta_1 b_1(X) + \dots + \beta_K b_K(X)$$

Examples of basis functions:

- *Polynomials:* $b_1(X) = X$, $b_2(X) = X^2, \dots$
 - Smooth but global
- *Step functions:* $b_1(X) = I(c_1 < X \leq c_2)$, $b_2(X) = I(c_2 < X \leq c_3), \dots$
 - Local but discontinuous
- *Splines:* piecewise polynomials with continuity constraints
 - Best of both: local flexibility like step functions, but smoother like polynomials

Main benefits:

- Nonlinear in X : the basis functions can capture nonlinearities more flexibly
- **Important distinction:** while these models may be nonlinear as functions of X , they are *linear in the unknown coefficients* β_k , so least squares still applies

Pop-up quiz: Basis functions

Suppose we fit

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 I(X > 50) + \varepsilon.$$

Question: Which statement is most accurate?

- A) The model must be fit separately for $X \leq 50$ and $X > 50$.
- B) The fitted regression function is necessarily continuous at $X = 50$.
- C) Least squares cannot be used because the model includes nonlinear functions of X .
- D) Least squares can be used after creating the transformed predictors X^2 and $I(X > 50)$, because the model is linear in the coefficients.

Answer: D. The fitted function is nonlinear in X , but linear in the unknown coefficients; we can treat X^2 and $I(X > 50)$ as transformed features and fit by least squares.

Regression splines: Main concept

Want: More flexible than a single polynomial, but smoother than step functions

Idea:

- *Piecewise polynomials* of degree d , joined at *knots* (cutpoints)
- Within each interval between knots, fit a separate polynomial (e.g. cubic)
- Impose *smoothness constraints* at each knot, preventing abrupt jumps or kinks

Example (degree $d = 3$, one knot at c): $Y = f(X) + \epsilon$ where

$$f(x) = \begin{cases} \beta_{01} + \beta_{11}x + \cdots + \beta_{d1}x^d, & x \leq c, \\ \beta_{02} + \beta_{12}x + \cdots + \beta_{d2}x^d, & x > c. \end{cases}$$

For a standard degree- d regression spline, we require f and its first $d - 1$ derivatives to be continuous at each knot

→ See the next slide for an illustration

Illustration of cubic spline: Ensuring smoothness at knots

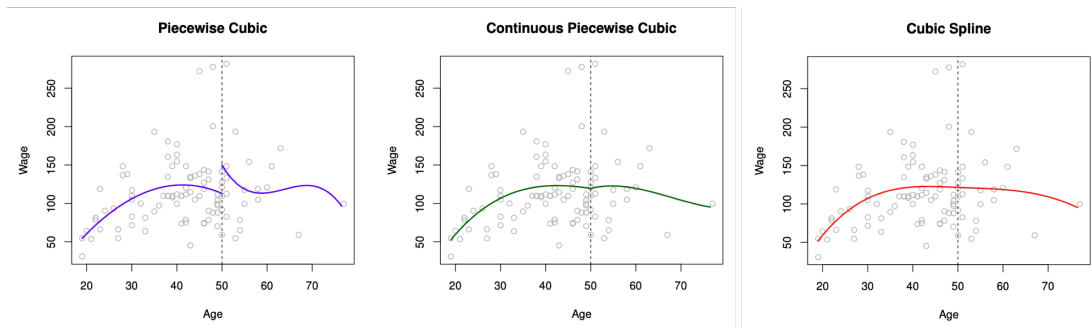


Figure: Various piecewise polynomials fit to a subset of the **Wage** data, with a knot at **age=50**. **Left:** Unconstrained polynomials cause a discontinuity. **Center:** Imposing continuity at the knot eliminates jumps but can form a “kink.” **Right:** Additionally constraining continuity of first and second derivatives yields a smooth cubic spline. Excerpted from [JWHT21, Figure 7.3].

Illustration: Comparing cubic vs. linear splines

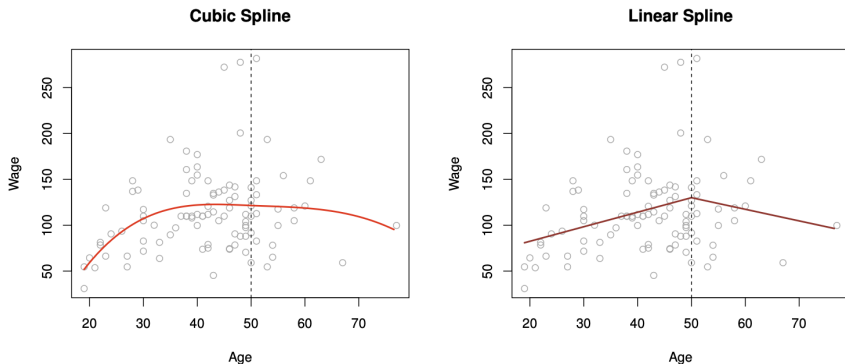


Figure: Again with a knot at **age=50** on a subset of **Wage**. **Left:** A cubic spline, enforcing continuity of function plus its 1st and 2nd derivatives. **Right:** A linear spline, only requiring continuity of function at the knot **age=50**. Excerpted from [JWHT21, Figure 7.3].

Degrees of freedom for splines

Degrees of freedom measure the effective number of fitted coefficients, hence the flexibility of the spline model

Piecewise polynomial of degree d with K knots:

- $(d + 1)$ polynomial coefficients in each of the $(K + 1)$ intervals
 $\implies (K + 1) \times (d + 1)$ parameters in total *before* smoothness constraints
- Each knot imposes d constraints: continuity of f and its first $d - 1$ derivatives
 - $K \times d$ constraints in total
- After smoothness constraints, the degrees of freedom are

$$(K + 1)(d + 1) - Kd = d + 1 + K.$$

- e.g., a cubic spline ($d = 3$) with K knots has $(3 + 1) + K = K + 4$ parameters

Trade-off:

- Enough degree and knots to capture possible nonlinearities
- But not so many that we overfit or lose interpretability

Spline basis representation: Truncated power basis

Key question: How to systematically fit a piecewise polynomial, enforcing the smoothness constraints at the knots?

Truncated power basis for a degree- d spline:

$$\underbrace{1, X, X^2, \dots, X^d}_{\text{base polynomials}} \cup \left\{ \underbrace{(X - c_k)_+^d}_{\text{truncated power basis}} : k = 1 \dots K \right\}$$

where $(x - c)_+^d = \max\{x - c, 0\}^d$

- Then, we can write

$$f(x) = \beta_0 + \beta_1 x + \dots + \beta_d x^d + \sum_{k=1}^K \beta_{d+k} (x - c_k)_+^d$$

- The term $(x - c_k)_+^d$ is inactive before the knot and changes the polynomial shape after it
- This representation **automatically** gives a degree- d spline with the desired smoothness

A toy numerical example: Piecewise linear spline

Example

Let X range in $[0, 8]$ with knots at $x = 2, 5$. Use piecewise linear polynomials (degree $d = 1$). By the DoF formula $(d + 1) + K = (1 + 1) + 2 = 4$ total parameters.

Basis representation:

$$b_0(x) = 1, \quad b_1(x) = x, \quad b_2(x) = (x - 2)_+, \quad b_3(x) = (x - 5)_+, \quad (u)_+ = \max(u, 0).$$

Then the resulting linear spline model, fit by least squares, is

$$\hat{y}(x) = \beta_0 b_0(x) + \beta_1 b_1(x) + \beta_2 b_2(x) + \beta_3 b_3(x).$$

Interpretation:

- β_0 is the base intercept.
- β_1 is the slope for $0 \leq x \leq 2$.
- β_2 changes the slope after $x = 2$, so the slope for $2 < x \leq 5$ is $\beta_1 + \beta_2$.
- β_3 further changes the slope after $x = 5$, so the slope for $5 < x \leq 8$ is $\beta_1 + \beta_2 + \beta_3$.

Pop-up quiz: Regression spline

Suppose we fit a **cubic spline** with $K = 3$ knots.

Question: How many degrees of freedom does this spline have after imposing the usual smoothness constraints?

- A) 4, because a cubic polynomial has four coefficients.
- B) 3, because there are three knots.
- C) $K + 4 = 7$, because a cubic spline has $4 + K$ degrees of freedom.
- D) $4(K + 1) = 16$, because there are four polynomial pieces and each has four coefficients.

Answer: C. Before constraints, there are $4(K + 1) = 16$ coefficients. The $K = 3$ knots impose $3K = 9$ smoothness constraints, leaving $16 - 9 = 7 = K + 4$ degrees of freedom.

Preview: Why natural splines?

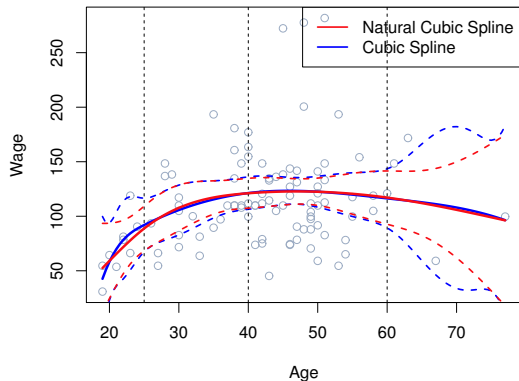


Figure: A cubic spline and a natural cubic spline, with three knots, fit to a subset of the `Wage` data. The dashed lines denote the knot locations [JWHT21, Figure 7.4].

Preview for next lecture:

- Cubic splines can behave erratically near the boundaries.
- Natural cubic splines add boundary constraints, making the tails linear.
- We will also discuss knot placement and smoothing splines.

Wrap-up & next steps

- **Basis functions** create transformed features $b_1(X), \dots, b_K(X)$, allowing nonlinear fits while still using least squares
- **Regression splines** use piecewise polynomials joined smoothly at knots
- **Truncated power bases** give a compact way to represent splines and fit them using ordinary least squares
- **Next lecture:** natural splines, knot placement, and smoothing splines

Suggested reading: [JWHT21, Ch. 7.1–7.4]

References



Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.

An Introduction to Statistical Learning: with Applications in R, volume 112 of *Springer Texts in Statistics*.

Springer, New York, NY, 2nd edition, 2021.